

HANKO TULLINIEMI, FinlandWMO: **027460**Lat: **59.809N**Lon: **22.916E**Elev: **90**StdP: **14.65**Time Zone: **2.00 (EUE)**Period: **09-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	3.7	8.7	-0.2	5.4	4.5	4.4	6.9	10.2	45.0	31.9	41.5	33.2	13.8	50	0.873

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.3	75.1	68.8	72.5	66.9	70.1	64.7	69.7	73.9	67.7	71.8	65.5	69.5	15.0	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.9	102.8	72.8	66.0	96.1	71.0	63.7	88.5	68.4	33.7	74.0	32.2	71.9	30.4	69.4	73.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
39.5	35.7	32.3	2.5	77.0	7.1	4.7	-2.6	80.4	-6.8	83.2	-10.7	85.9	-15.9	89.3		
			DB													
			WB	1.9	69.9	6.9	3.5	-3.1	72.4	-7.1	74.5	-11.0	76.4	-16.1	78.9	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	44.6	27.0	27.3	31.1	38.6	49.2	56.9	64.3	63.5	56.8	46.2	39.9	33.7
	DBStd	14.39	8.16	8.97	6.10	4.41	6.25	4.21	5.05	3.76	4.58	5.50	6.50	7.74
	HDD50	3330	712	635	585	344	94	2	0	0	8	139	304	507
	HDD65	7515	1177	1055	1050	793	490	243	76	74	247	584	754	972
	CDD50	1373	0	0	0	1	69	210	444	417	212	20	0	0
	CDD65	83	0	0	0	0	0	2	55	26	0	0	0	0
	CDH74	119	0	0	0	0	0	2	109	8	0	0	0	0
	CDH80	2	0	0	0	0	0	0	2	0	0	0	0	0
Wind	WSAvg	17.0	17.7	17.5	17.6	15.6	15.3	15.4	14.2	14.7	16.8	18.3	19.8	21.4
Precipitation	PrecAvg	25.9	2.4	1.7	1.6	1.4	1.1	1.6	2.2	3.0	2.2	3.0	3.1	2.6
	PrecMax	30.0	4.3	3.9	3.7	2.5	2.4	3.0	6.5	6.9	5.7	7.0	6.4	4.7
	PrecMin	18.8	0.3	0.1	0.2	0.4	0.2	0.6	0.0	0.3	0.6	1.2	0.3	0.7
	PrecStd	3.2	1.1	1.0	0.7	0.6	0.6	0.7	1.7	1.6	1.3	1.4	1.5	1.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.9	39.7	43.5	54.9	67.0	69.9	80.0	75.6	66.7	57.5	50.7	46.6
		MCWB	39.4	36.6	38.0	46.5	56.5	62.4	69.3	68.9	61.8	54.5	49.6	44.9
	2%	DB	39.6	37.5	40.5	50.0	63.9	67.2	77.1	72.7	65.0	55.7	49.5	45.0
		MCWB	38.3	36.0	36.8	44.1	55.3	60.0	68.8	67.2	60.8	53.4	48.4	43.3
	5%	DB	38.2	36.6	38.9	46.9	61.1	65.0	74.6	70.8	63.4	54.5	48.1	43.8
		MCWB	37.0	35.4	36.0	42.0	54.4	58.7	68.5	65.4	59.7	52.5	46.7	42.0
	10%	DB	36.6	35.7	37.6	44.4	58.5	63.1	72.2	68.8	62.1	53.2	46.9	42.4
		MCWB	35.5	34.8	35.5	40.5	53.0	57.8	66.8	63.9	58.8	51.2	45.3	40.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	40.0	37.6	39.1	47.1	58.8	64.2	71.9	70.2	63.6	55.6	50.0	45.7
		MCDB	40.6	38.4	41.7	54.0	64.8	68.3	76.3	73.5	66.0	56.5	50.2	46.2
	2%	WB	38.6	36.6	37.7	44.4	56.8	61.8	70.7	68.2	61.6	54.3	48.8	43.8
		MCDB	39.4	37.3	39.5	49.1	61.9	66.0	75.1	71.8	63.7	55.3	49.4	44.6
	5%	WB	37.3	35.8	36.9	42.6	55.1	60.1	69.3	66.7	60.5	53.1	47.3	42.3
		MCDB	38.1	36.4	38.3	46.1	60.0	63.7	73.7	70.0	62.4	54.1	48.0	43.3
	10%	WB	35.8	35.0	35.9	41.0	53.4	58.7	67.4	64.8	59.6	51.5	45.7	40.8
		MCDB	36.5	35.6	37.1	43.7	57.8	62.1	71.8	68.0	61.6	52.8	46.6	42.2
Mean Daily Temperature Range	5% DB	MDBR	7.0	5.9	7.0	7.0	7.9	7.1	7.3	6.3	5.6	5.4	4.7	5.6
		MCDBR	4.9	4.4	6.5	10.5	11.4	9.8	8.8	7.3	6.4	5.1	4.7	4.8
		MCWBR	5.1	3.9	4.7	6.4	6.9	5.6	5.4	4.9	4.5	4.9	5.1	5.3
	5% WB	MCDBR	4.7	4.2	5.8	9.4	11.1	8.4	8.2	6.1	5.5	4.6	4.5	4.9
		MCWBR	5.0	4.0	4.5	5.9	7.3	5.5	5.5	5.3	4.3	5.1	5.1	5.6
Clear-Sky Solar Irradiance	taub	0.281	0.313	0.333	0.371	0.366	0.362	0.389	0.382	0.351	0.334	0.311	0.267	
	taud	2.168	2.235	2.300	2.325	2.422	2.476	2.421	2.432	2.498	2.407	2.218	2.099	
	Ebn at Noon	155	216	252	263	273	277	265	257	247	210	141	111	
	Edh at Noon	14	23	29	33	33	31	33	30	24	19	13	10	
All-Sky Solar Radiation	RadAvg	77	243	727	1256	1760	1859	1754	1382	841	365	99	49	
	RadStd	12	44	86	131	172	119	148	136	103	49	19	6	